

Nộp bài tập homework ss07

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Lớp	PD16

- Highlight 1 worst reg2reg setup timing path and 1 worst mem2reg timing path in design.
- Path reg2reg: **report_timing -path_group reg2reg**

```

innovus 26> report_timing -path_group reg2reg
#####
# Generated by: Cadence Innovus 20.28-pd06.1
# OS: Linux x86_64(host ID tctc-eda-be-9-ldap-1)
# Generated on: Thu Jul 30 21:28:35 2026
# Design: croc_chip
# Command: report_timing -path_group reg2reg
#####
Path 1: VIOLATED Setup Check with Pin 1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_prefetch_buffer_1_fifo_1_rdata_q_68_reg/CLK
Endpoint: 1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_prefetch_buffer_1_fifo_1_rdata_q_68_reg/D (v) checked with Loading edge of 'clk_sys'
Beginpoint: 1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/Q (v) triggered by Loading edge of 'clk_sys'
Path Groups: (reg2reg)
Analysis View: func_view.uc
Other End Arrival Time 0.000
+ Setup 0.130
+ Phase Shift 10.000
+ Uncertainty 0.100
+ Required Time 0.764
+ Arrival Time 0.766
+ Slack Time -0.002
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) -0.234
+ Beginpoint Arrival Time -0.234
Pin Cell Fanout Load Slew Delay Arrival Pin
Time Location
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/CLK - 4983 0.000 0.200 - -0.234 (1140.95, 676.02)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/Q sg13g2_dfrbp_2 7 0.038 0.110 0.485 0.251 (1134.24, 676.26)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/D sg13g2_mmr2_2 3 0.014 0.193 0.238 0.409 (1111.29, 698.88)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/V sg13g2_mmr2_1 2 0.009 0.140 0.244 0.753 (1095.36, 715.68)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/W sg13g2_d220i_1 2 0.008 0.150 0.194 0.947 (1096.24, 725.75)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/X sg13g2_buf_4 8 0.014 0.045 0.176 1.123 (1085.76, 727.44)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/Y sg13g2_or2_1 1 0.008 0.000 0.190 1.313 (1089.60, 744.24)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/Z sg13g2_and2b_1 1 0.008 0.100 0.171 1.484 (1126.08, 746.33)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/0 sg13g2_and2_1 1 0.010 0.000 0.002 1.566 (1104.40, 745.92)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/1 sg13g2_and2_2 2 0.008 0.058 0.087 1.683 (1085.76, 746.66)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/2 sg13g2_d220i_1 1 0.010 0.201 0.199 1.762 (1073.28, 747.58)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/3 sg13g2_d220i_1 1 0.003 0.224 0.157 1.920 (1046.48, 757.68)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/4 sg13g2_buf_3 2 0.013 0.054 0.172 2.103 (1043.04, 759.36)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/5 sg13g2_tnv_2 2 0.012 0.059 0.085 2.187 (1045.44, 759.36)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/6 sg13g2_and2_2 2 0.008 0.071 0.085 2.261 (1043.04, 763.56)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/7 sg13g2_and2_2 2 0.024 0.068 0.196 2.358 (1047.36, 761.64)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/8 sg13g2_and2_1 1 0.010 0.079 0.198 2.465 (984.48, 791.28)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/9 sg13g2_tnv_1 2 0.010 0.052 0.071 2.530 (956.96, 789.60)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/10 sg13g2_tnv_2 1 0.000 0.030 0.037 2.583 (959.84, 795.48)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/11 sg13g2_mmr2_2 1 0.007 0.077 0.093 2.674 (956.16, 790.48)
1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/12 sg13g2_and2_1 1 0.008 0.053 0.077 2.753 (932.48, 791.28)

```

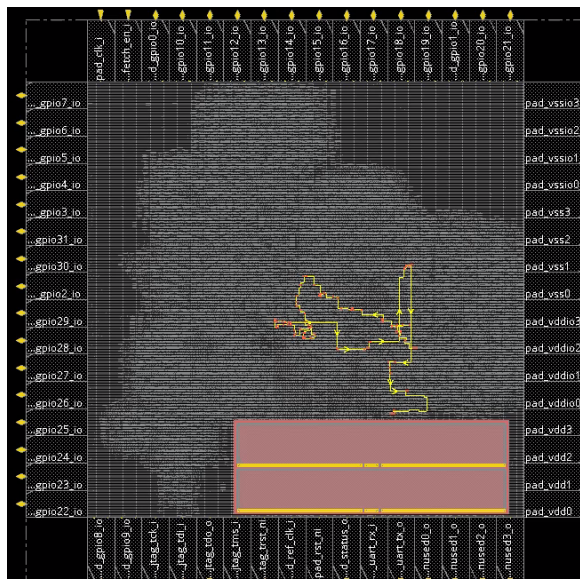
- Lấy begin point và end point để report timing và highlight path này lên

```

innovus 27> report_timing -from 1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_controller_1_instr_1_5_reg/Q -to 1_croc_soc/i_croc/i_core_wrap/i_tbox_if_stage_1_prefetch_buffer_1_fifo_1_rdata_q_68_reg/D -machine_readable > temp_rpt
innovus 28> load
load loadBlackBoxNetlist loadConfig2 loadDefile loadDrc loadECO loadFPlan loadFootPrint loadIoFile loadLeFile loadPtnPin loadSpectralRoute loadViolationReport loadWorkspace load_eco_models
innovus 28> load_timing_debug_report temp_rpt
Parsing file temp_rpt...
innovus 29> high
highlight highlight_pin highlight_pin_connection highlight_timing_report
innovus 29> highlight_timing_report -path 1
0
innovus 30> win
innovus 31>

```

- Kết quả highlight path reg2reg



○ Path timing mem2reg: **report_timing -path_group mem2reg**

```

Innovus 32> report_timing -path_group mem2reg
=====
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (host ID fctc-cda-be-9-ldap-1)
# Generated on: Thu Jul 30 21:36:05 2020
# Design: croc_chtp
# Command: report_timing -path_group mem2reg
=====
Path 1: MET Setup Check with Pin f_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_20_reg/CLK
Endpoint: f_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_20_reg/D (v) checked with leading edge of 'clk_sys'
Beginpoint: f_croc_soc/i_croc/gen_sram_bank_1_1_sram/gen_512x32x8x1_1_cut/A_DOUT[59] (v) triggered by leading edge of 'clk_sys'
Path Groups: {mem2reg}
Analysis View: func_view_wc
Other End Arrival Time
- Setup 0.171
+ Phase Shift 10.000
- Uncertainty 0.100
+ Required Time 9.585
- Arrival Time 9.485
+ Slack Time 0.000
+ Clock Rise Edge 0.000
+ Clock Network Latency (ideal) -0.037
- Beginpoint Arrival Time -0.037
-----
Pin Cell Fanout Load Slew Delay Arrival Pin
Time Location
-----
f_croc_soc/i_croc/gen_sram_bank_1_1_sram/gen_512x32x8x1_1_cut/A_CLK - 4983 0.000 0.200 - -0.037 (1006.00, 202.19)
f_croc_soc/i_croc/gen_sram_bank_1_1_sram/gen_512x32x8x1_1_cut/A_DOUT[59] RH_IHP5G13_1P_250x64_c2_ba_bist 1 0.005 0.055 5.118 5.079 (1346.84, 202.19)
f_croc_soc/i_croc/gen_sram_bank_1_1_sram/gen_512x32x8x1_1_cut/A_DOUT[59] sg13g2_mux2_1 1 0.023 0.149 0.200 0.309 (1351.20, 189.00)
f_croc_soc/i_croc/ictc_preCTS_FE_OF02744_all_sbr_obi_rsp_153/x sg13g2_buf_4 3 0.112 0.166 0.264 5.623 (1417.44, 267.96)
f_croc_soc/i_croc/_6077_/V sg13g2_a221oi_1 1 0.005 0.214 0.302 5.925 (1025.28, 502.32)
f_croc_soc/i_croc/_6078_/V sg13g2_nor2_1 1 0.005 0.097 0.114 6.089 (1024.00, 509.80)
f_croc_soc/i_croc/_6079_/V sg13g2_a21oi_1 1 0.004 0.109 0.112 6.101 (1025.28, 521.64)
f_croc_soc/i_croc/_6080_/V sg13g2_a21oi_1 4 0.014 0.256 0.152 6.314 (1025.28, 526.68)
f_croc_soc/i_croc/i_core_wrap/_7775_/V sg13g2_nor2b_1 1 0.004 0.090 0.198 6.511 (1078.56, 526.68)
f_croc_soc/i_croc/i_core_wrap/_7776_/V sg13g2_a21oi_1 3 0.020 0.209 0.270 6.708 (1079.32, 534.24)
f_croc_soc/i_croc/i_core_wrap/_7885_/V sg13g2_a21oi_1 1 0.005 0.125 0.203 6.993 (1115.52, 559.44)
f_croc_soc/i_croc/i_core_wrap/_7894_/V sg13g2_buf_4 28 0.134 0.194 0.273 7.266 (1114.08, 568.68)
f_croc_soc/i_croc/i_core_wrap/_7895_/V sg13g2_nor2_2 5 0.020 0.224 0.295 7.562 (1116.16, 570.84)
f_croc_soc/i_croc/i_core_wrap/_7895_/V sg13g2_nand2_1 6 0.039 0.307 0.418 7.979 (1127.52, 613.62)
f_croc_soc/i_croc/i_core_wrap/_7896_/V sg13g2_nor2_1 1 0.005 0.151 0.191 8.178 (1144.80, 615.72)
f_croc_soc/i_croc/i_core_wrap/ictc_preCTS_FE_OF01847_3001/x sg13g2_buf_1 6 0.035 0.221 0.239 8.455 (1148.05, 623.36)
f_croc_soc/i_croc/i_core_wrap/_8865_/V sg13g2_nand3_1 1 0.010 0.202 0.240 8.703 (1173.12, 627.48)
f_croc_soc/i_croc/i_core_wrap/_8868_/V sg13g2_nand3_1 1 0.005 0.088 0.123 8.828 (1181.28, 594.72)
f_croc_soc/i_croc/i_core_wrap/_8872_/V sg13g2_a21oi_1 1 0.005 0.123 0.128 8.957 (1180.32, 589.68)
f_croc_soc/i_croc/i_core_wrap/_8873_/V sg13g2_a21oi_1 1 0.004 0.128 0.098 9.054 (1177.44, 590.92)
f_croc_soc/i_croc/i_core_wrap/ictc_preCTS_FE_OF06570_3107/x sg13g2_buf_1 1 0.025 0.164 0.234 9.289 (1179.09, 608.24)

```

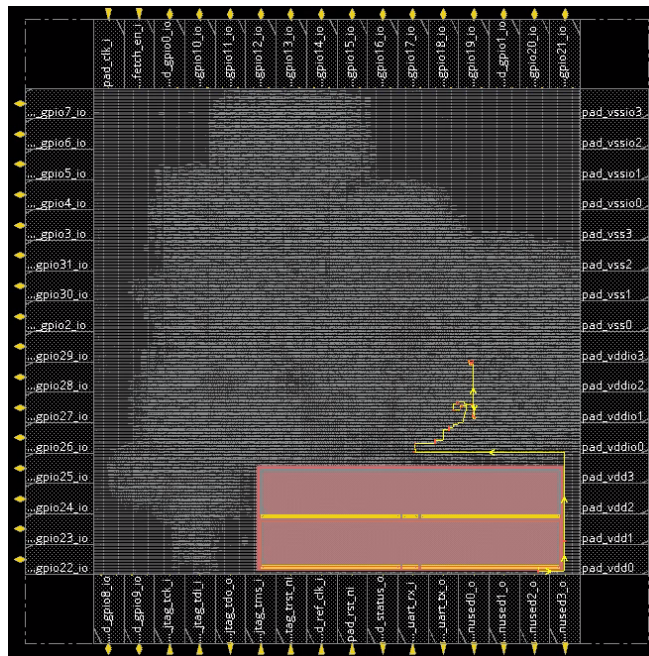
○ Tương tự như trên vẫn lấy begin và end point để highlight path này lên

```

Innovus 33> report_timing -from f_croc_soc/i_croc/gen_sram_bank_1_1_sram/gen_512x32x8x1_1_cut/A_DOUT -to f_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_20_reg/D -machine_readable > temp_rpt
Innovus 34> load_timing_debug_report temp_rpt
Parsing file temp_rpt...
Innovus 35> highlight_timing_report -path 1
0
Innovus 36> win
Innovus 37>

```

○ Kết quả highlight



- Check design utilization and compare this number with placeOnly step? Can you list some reason is changed (inscrease/descrease) from placeOnly to finish placement step?

- Report ulti ở giai đoạn **place_opt**

```
innovus 38> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=2945.5M, init mem=2947.0M)
Pre-route DRC Violation:      285
*info: Placed = 46640          (Fixed = 10253)
*info: Unplaced = 0
Placement Density:43.15%(600380/1391267)
Placement Density (including fixed std cells):44.01%(621775/1412663)
Finished checkPlace (total: cpu=0:00:03.0, real=0:00:01.0; vio checks: cpu=0:00:01.6, real=0:00:01.0; mem=2947.0M)
1
innovus 39> █
```

- Report ulti ở giai đoạn **place_only**

```
innovus 44> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=2510.7M, init mem=2510.7M)
Pre-route DRC Violation:      228
*info: Placed = 40240          (Fixed = 10253)
*info: Unplaced = 0
Placement Density:38.98%(542379/1391267)
Placement Density (including fixed std cells):39.91%(563774/1412663)
Finished checkPlace (total: cpu=0:00:02.6, real=0:00:01.0; vio checks: cpu=0:00:01.2, real=0:00:00.0; mem=2510.7M)
1
innovus 45> █
```

- Từ 2 report Ulti trên (checkPlace) ta có thể thấy được ở bước cuối của placement (place_opt) đã tăng khoảng 5% so với giai đoạn đầu của placement (place_only). Cụ thể ở các cell Placed của place only chỉ có 40 240 cell khi ở cuối giai đoạn thì placed có 46 640 cell, tăng khoảng 6000 cell. Các nguyên nhân dẫn đến việc tăng diện tích được liệt kê dưới đây.
 - Insert thêm Buffer/Inverter để fix timing, transition
 - Inser thêm Buffer để split fanout
 - Upsize cell lớn hơn để fix timing, có drive strength lớn hơn -> diện tích lớn hơn.

Homework ss07 Extension

01. Please open design database (launch innovus + source below dir to open database).

Source ictc/student_data/share/pd/07_ss7/HW/02_place_opt.invs

02. Basic proc in tcl: proc proc_name {argument1 argument2 ... argument_n} {
body_script }

- Proc sum number:
 - Sum 2 number:

```
proc add_number {a b} {  
    set sum [expr {$a + $b}]  
    return $sum  
}  
  
innovus 5> source scripts/sum_2_num.tcl  
innovus 6>  
innovus 6> puts [add_number 5 7]  
12
```

- Sum many numbers

```
proc sum_numbers {args} {  
    set total 0  
    foreach num $args {  
        set total [expr {$total + $num}]  
    }  
    return $total  
}  
  
innovus 9> source scripts/sum_many_num.tcl  
innovus 10> puts [sum]  
sum_numbers summaryReport  
innovus 10> puts [sum_numbers 1 2 3 4 5]  
15  
innovus 11>
```

- Proc get std cell info:

```
proc getInstInfo {args} {  
    set n 0  
    foreach inst $args {  
        if {[dbGetInstByName $inst] == "0x0"} {  
            puts "#INFO: input inst {$inst} is not existed!"  
            continue  
        }  
        set cell_type [dbget [dbGetInstByName $inst].cell.name]  
        set status [dbget [dbGetInstByName $inst].pstatus]  
        set area [dbget [dbGetInstByName $inst].area]  
        set location [dbget [dbGetInstByName $inst].pt]  
        incr n;  
        puts "#$n $inst $cell_type $status $area $location"  
    }  
}  
  
innovus 22> source scripts/get_cell_infor.tcl  
innovus 23> getInstInfo i_croc_soc/i_croc/i_timer/counter_hi_i.counter_value_o_22_reg i_croc_soc/i_croc/i_timer/counter_hi_i.counter_value_o_18_reg  
#1 i_croc_soc/i_croc/i_timer/counter_hi_i.counter_value_o_22_reg sgl3g2_dfrbp_1 placed 47.1744 {734.4 955.08}  
#2 i_croc_soc/i_croc/i_timer/counter_hi_i.counter_value_o_18_reg sgl3g2_dfrbp_1 placed 47.1744 {720.36 951.3}  
innovus 24>
```

03. Check total area of standard cell type:

- Script

```
proc getInstAreaByCellName {args} {  
    set n 0  
    foreach inst $args {  
        if {[dbget top.insts.cell.name $inst -p2] == "0x0"} {  
            puts "#INFO: input inst {$inst} is not existed!"  
            continue  
        }  
        set area_list [dbget [dbget top.insts.cell.name $inst -p2].area]  
  
        set total_area [expr [join $area_list "+"]]  
        incr n  
        puts "# $n $inst $total_area"  
    }  
}
```

- Kết quả

```
innovus 32> source scripts/total_area_cell.tcl  
innovus 33> getInstAreaByCellName sgl3g2_dfrbp_1  
# 1 sgl3g2_dfrbp_1 246391.8912  
innovus 34> win  
innovus 35> dbget selected.cell.name  
sgl3g2_dfrbp_1  
innovus 36> win  
innovus 37> dbget selected.cell.name  
sgl3g2_a2loi_1  
innovus 38> getInstAreaByCellName sgl3g2_dfrbp_1 sgl3g2_a2loi_1 sgl3g2_a2loi_3  
# 1 sgl3g2_dfrbp_1 246391.8912  
# 2 sgl3g2_a2loi_1 19477.584  
#INFO: input inst {sgl3g2_a2loi_3} is not existed!  
innovus 39>
```

04. Write proc auto query size cell by control up/down + level.

- Script:

```
1 proc getAutoCellSize {inst control size} {  
2     #check exist inst  
3     set inst_ptr [dbGetInstByName $inst]  
4  
5     if {$inst_ptr == "0x0"} {  
6         puts "#INFO: input instance {$inst} is not existed!"  
7         return  
8     }  
9  
10    # Check control  
11    set control [string tolower $control]  
12  
13    if {$control ne "up" && $control ne "down"} {  
14        puts "$ERROR: control must to be up or down!"  
15        puts "Usage: getAutoCellSize <inst> <up/down> <size>"  
16        return  
17    }  
18  
19    # Take current size cell  
20    set current_cell [dbget $inst_ptr.cell.name]  
21  
22    # cut string driver strength final  
23    set last_score [string last "_" $current_cell]  
24  
25    set cell_prefix [string range $current_cell 0 $last_score]  
26  
27    # find cell same function but diff size  
28    set avai_list [lsort -dictionary [dbget head.libCells.name "{$cell_prefix}*"]]  
29  
30    if {$avai_list == "0x0" || [llength $avai_list] == 0} {  
31        puts "$ERROR: cannot find available cell list for {$current_cell}!"  
32        return  
33    }  
34  
35    puts "INFO: Available list {$avai_list}"  
36  
37    # if only have 1 cell  
38    if {[llength $avai_list] == 1} {  
39        puts "#INFO: current cell size {$current_cell} is minimum! No size needed!"  
40        return  
41    }  
42  
43    # find position current cell in available list  
44    set current_index [lsearch -exact $avai_list $current_cell]  
45  
46    set last_index [expr {[llength $avai_list] - 1}]  
47  
48    # calculate new index  
49    set new_index [expr {[llength $avai_list] - 1}]  
50  
51    # calculate new index  
52    if {$control eq "up"} {  
53        set new_index [expr {$current_index + $size}]  
54  
55        if {$new_index > $last_index} {  
56            set new_index $last_index  
57        }  
58    } else {  
59        set new_index [expr {$current_index - $size}]  
60        if {$new_index < 0} {  
61            set new_index 0  
62        }  
63    }  
64  
65    # take cell type  
66    set target_cell [lindex $avai_list $new_index]  
67  
68    set control_name "[string toupper [string index $control 0]][string range $control 1 end]"  
69  
70    puts "#INFO: $inst - CurrentSize: $current_cell - {$control_name}_{$size}_Level: $target_cell"  
71    return $target_cell  
72 }
```

- Kết quả

```

innovus 78> getAutoCellSize i_croc_soc/i_croc/i_core_wrap/FE_RC_9456_0 up 2
INFO: Available list {sg13g2_inv_1 sg13g2_inv_2 sg13g2_inv_4 sg13g2_inv_8 sg13g2_inv_16}
#INFO: i_croc_soc/i_croc/i_core_wrap/FE_RC_9456_0 - CurrentSize: sg13g2_inv_1 - Up_2_Level: sg13g2_inv_4
sg13g2_inv_4
innovus 79> getAutoCellSize i_croc_soc/i_croc/i_core_wrap/FE_RC_9456_0 down 4
INFO: Available list {sg13g2_inv_1 sg13g2_inv_2 sg13g2_inv_4 sg13g2_inv_8 sg13g2_inv_16}
#INFO: i_croc_soc/i_croc/i_core_wrap/FE_RC_9456_0 - CurrentSize: sg13g2_inv_1 - Down_4_Level: sg13g2_inv_1
sg13g2_inv_1
innovus 80> getAutoCellSize i_croc_soc/i_croc/i_core_wrap/FE_RC_9456_0 up 4
INFO: Available list {sg13g2_inv_1 sg13g2_inv_2 sg13g2_inv_4 sg13g2_inv_8 sg13g2_inv_16}
#INFO: i_croc_soc/i_croc/i_core_wrap/FE_RC_9456_0 - CurrentSize: sg13g2_inv_1 - Up_4_Level: sg13g2_inv_16
sg13g2_inv_16
innovus 81> getAutoCellSize i_croc_soc/i_croc/i_core_wrap/FE_RC_9456 up 4
#INFO: input instance {i_croc_soc/i_croc/i_core_wrap/FE_RC_9456} is not existed!
innovus 82>

```

05. Write proc to summary info output from report_timing command.

* Script

```

1 proc summaryTiming {{group reg2reg}} (max_paths 1) {
2     set_table_style -nosplit -no_frame_fix_width
3
4     set rpt "timing_${group}.rpt"
5     redirect $rpt {
6         report_timing -path_group $group -max_paths $max_paths
7     }
8
9     set fb [open $rpt r]
10    set idx 0
11
12    set start ""
13    set end ""
14    set slack ""
15
16    set total_level 0
17    set buf_num 0
18    set logic_num 0
19
20    set max_delay 0.0
21    set max_slew 0.0
22
23    while {[gets $fb line] >= 0} {
24        if {[regexp {"^Path[:][[:space:]]+(\S+)} $line -> path_no}] {
25            if {$idx != 0} {
26                puts "$idx $start $end $total_level $buf_num $logic_num $max_slew $max_delay $slack"
27            }
28            set idx $path_no
29
30            set start ""
31            set end ""
32            set slack "N/A"
33
34            set total_level 0
35            set buf_num 0
36            set logic_num 0
37
38            set max_delay 0
39            set max_slew 0
40            continue
41        }
42
43        if {[regexp {"^Beginpoint[:][[:space:]]+(\S+)} $line -> start_pin}] {
44            set start $start_pin
45            regsub {[/\s]} $start "" start
46            continue
47        }
48        if {[regexp {"^Endpoint[:][[:space:]]+(\S+)} $line -> end_pin}] {
49            set end $end_pin
50            regsub {[/\s]} $end "" end
51            continue
52        }
53
54        if {[regexp {"^Slack Time[:][[:space:]]+(\S+)} $line -> slack_value}] {
55            set slack $slack_value
56            continue
57        }
58
59        set tok [regexp -inline -all {\S+} $line]
60
61        set cell_index [lsearch -glob $tok "sg13g2_*"]
62
63        if {$cell_index >= 0} {
64            set cell [lindex $tok $cell_index]
65            set delay [lindex $tok [expr {$cell_index + 1}]]
66            set slew [lindex $tok [expr {$cell_index + 2}]]
67
68            if {[string is double -strict $delay] && [string is double -strict $slew]} {
69                incr total_level
70
71                incr buf_num
72
73                if {[regexp -nocase {_(buf|inv)_} $cell]} {
74                    } else {
75                        incr logic_num
76                    }
77
78                if {$delay > $max_delay} {
79                    set max_delay $delay
80                }
81
82                if {$slew > $max_slew} {
83                    set max_slew $slew
84                }
85            }
86        }
87
88        if {$idx != 0} {
89            puts "$idx $start $end $total_level $buf_num $logic_num $max_slew $max_delay $slack"
90        }
91    }
92    close $fb
93}

```

- Trích xuất thông tin report timing 1 path

```

innovus 114> summaryTiming reg2reg
#1 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_34__reg 84 22 62 0.216 0.518 -0.007
innovus 115>

```

- Trích xuất thông tin report timing 7 paths

```

innovus 113> summaryTiming reg2reg 7
#1 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_34__reg 84 22 62 0.216 0.518 -0.007
#2 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.crash_dump_o_82__reg 86 22 64 0.216 0.518 -0.007
#3 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_36__reg 84 22 62 0.216 0.518 -0.004
#4 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_61__reg 84 21 63 0.216 0.518 -0.004
#5 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_57__reg 84 22 62 0.216 0.518 -0.003
#6 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_55__reg 84 22 62 0.216 0.518 -0.003
#7 i_croc_soc/i_croc/i_core_wrap/i_ibex.id_stage_i.controller_i.instr_i_16__reg i_croc_soc/i_croc/i_core_wrap/i_ibex.if_stage_i.prefetch_buffer_i.fifo_i.rdata_q_56__reg 84 21 63 0.216 0.518 -0.002

```